

Elementary Mathematics: K–5 Foundations

This guide covers the core ideas normally developed from kindergarten through grade 5. It is a foundation map rather than a claim about any one school system's standards. Read the concepts in order; later ideas use earlier ones.

Complete concept outline

Number and operations

- **N1. Counting, place value, and comparing whole numbers**
- **N2. Addition and subtraction**
- **N3. Multiplication**
- **N4. Division and remainders**
- **N5. Factors, multiples, primes, and divisibility**
- **N6. Order of operations and equations**
- **N7. Estimation, mental math, and problem-solving strategies**

Fractions, decimals, and relationships

- **N8. Fraction meaning and equivalence**
- **N9. Comparing and operating with fractions**
- **N10. Decimal place value and operations**
- **N11. Ratios, rates, and percentages**
- **N12. Patterns and introductory variables**

Measurement and geometry

- **N13. Measurement and unit conversion**
- **N14. Time and money**
- **N15. Shapes, lines, angles, and symmetry**
- **N16. Perimeter, area, and volume**
- **N17. Coordinates and transformations**

Data and chance

- **N18. Tables, graphs, and data questions**
- **N19. Mean, median, mode, and range**
- **N20. Probability and mathematical communication**

Worked examples

N1. Counting, place value, and comparing whole numbers

Place value gives each digit a value based on its position: ones, tens, hundreds, and so on. Compare the greatest place first; rounding replaces a number with a nearby, easier number.

1. **What is the value of 7 in 4,728?** Solution: It is in the hundreds place, so its value is **700**.

2. **Which is greater, 6,305 or 6,350?** Solution: Thousands and hundreds match; compare tens: 5 tens is greater than 0 tens, so **6,350 > 6,305**.
3. **Round 4,671 to the nearest hundred.** Solution: The tens digit is 7, so round 4,600 up to **4,700**.
4. **Write 8,042 in expanded form.** Solution: **8,000 + 40 + 2** (there are zero hundreds).
5. **What number is 100 more than 2,999?** Solution: Adding one hundred changes the hundreds place and carries: $2,999 + 100 = 3,099$.

N2. Addition and subtraction

Addition combines quantities. Subtraction finds a difference or removes a quantity. Use place value, regroup when a column reaches 10, and check inverse operations.

1. **347 + 286.** Solution: Ones $7 + 6 = 13$; write 3, carry 1. Tens $4 + 8 + 1 = 13$; write 3, carry 1. Hundreds $3 + 2 + 1 = 6$, so **633**.
2. **902 – 475.** Solution: Regroup across the zero: 902 becomes 8 hundreds, 9 tens, 12 ones. Subtract to get **427**.
3. **A farmer has 238 red apples and 167 green apples. How many apples?** Solution: $238 + 167 = 405$ apples.
4. **A book has 512 pages. Maya reads 289. How many remain?** Solution: $512 - 289 = 223$ pages.
5. **Find the missing number: $468 + \underline{\quad} = 700$.** Solution: Subtract 468 from 700: $700 - 468 = 232$.

N3. Multiplication

Multiplication represents equal groups, rectangular arrays, and repeated addition. The factors are multiplied to find a product; use place value for larger factors.

1. **6×4 .** Solution: Six groups of four: $4 + 4 + 4 + 4 + 4 + 4 = 24$.
2. **23×5 .** Solution: $(20 \times 5) + (3 \times 5) = 100 + 15 = 115$.
3. **34×12 .** Solution: $34 \times 10 = 340$ and $34 \times 2 = 68$; $340 + 68 = 408$.
4. **A theater has 8 rows of 27 seats. How many seats?** Solution: $8 \times 27 = 8 \times 20 + 8 \times 7 = 160 + 56 = 216$ seats.
5. **A box holds 9 packs of 12 stickers. How many stickers?** Solution: $9 \times 12 = 9 \times 10 + 9 \times 2 = 90 + 18 = 108$ stickers.

N4. Division and remainders

Division shares equally or counts equal groups. The dividend is divided by the divisor to find the quotient. A remainder is what is left and must be smaller than the divisor.

1. **$24 \div 6$.** Solution: Six groups fit into 24 four times, so the quotient is **4**.
2. **$73 \div 8$.** Solution: $8 \times 9 = 72$, leaving 1, so **9 remainder 1**.
3. **45 cookies shared equally among 5 children.** Solution: $45 \div 5 = 9$ cookies each.
4. **$146 \div 4$.** Solution: $4 \times 36 = 144$, leaving 2, so **36 remainder 2**.
5. **A bus seats 12 students. How many full buses for 50 students?** Solution: $50 \div 12 = 4$ remainder 2; two students still need a bus, so round up to **5 buses**.

N5. Factors, multiples, primes, and divisibility

A factor divides a number evenly. A multiple is a product of a number and a whole number. A prime has exactly two positive factors: 1 and itself.

1. **List the factors of 18.** Solution: Factor pairs are 1×18 , 2×9 , and 3×6 , so **1, 2, 3, 6, 9, 18**.
2. **Is 29 prime?** Solution: It is not divisible by 2, 3, 4, or 5; only 1 and 29 divide it, so **yes, 29 is prime**.
3. **Give the first five positive multiples of 7.** Solution: 7×1 through 7×5 gives **7, 14, 21, 28, 35**.
4. **Is 246 divisible by 3?** Solution: Its digit sum is $2 + 4 + 6 = 12$, and 12 is divisible by 3, so **yes**.
5. **Find the greatest common factor of 12 and 18.** Solution: Common factors are 1, 2, 3, and 6; the greatest is **6**.

N6. Order of operations and equations

Parentheses are handled first, then multiplication or division from left to right, then addition or subtraction from left to right. An equation is true when both sides have equal value.

1. **$7 + 3 \times 4$.** Solution: Multiply first: $3 \times 4 = 12$; $7 + 12 = \mathbf{19}$.
2. **$(7 + 3) \times 4$.** Solution: Parentheses first: $10 \times 4 = \mathbf{40}$.
3. **$36 \div 6 + 5$.** Solution: Divide first: $6 + 5 = \mathbf{11}$.
4. **Solve $5x = 35$.** Solution: Divide both sides by 5: $x = \mathbf{7}$.
5. **Is $18 - 4 = 3 \times 5$ true?** Solution: Left side is 14; right side is 15, so **no**. The equation is false.

N7. Estimation, mental math, and problem-solving strategies

Estimate to judge whether an answer is reasonable. Break numbers apart, use friendly numbers, draw a model, make a table, or work backward. State units and check the result.

1. **Estimate $398 + 205$.** Solution: Round to $400 + 200 = \mathbf{about\ 600}$; the exact sum is 603.
2. **Estimate 49×21 .** Solution: $50 \times 20 = \mathbf{about\ 1,000}$.
3. **Which is reasonable for $798 \div 8$: 10, 100, or 1,000?** Solution: $800 \div 8 = 100$, so **100** is reasonable.
4. **A \$20 bill buys items costing \$6.75 and \$8.10. Is \$5 enough change?** Solution: Total is \$14.85; change is $\$20 - \$14.85 = \mathbf{\$5.15}$, so yes.
5. **A rectangle has 4 equal sides and perimeter 28 cm. Find one side.** Solution: $28 \div 4 = \mathbf{7\ cm}$.
This uses the shape information before calculating.

N8. Fraction meaning and equivalence

A fraction names equal parts of a whole or collection. The denominator names the number of equal parts; the numerator counts selected parts. Equivalent fractions name the same amount.

1. **What does $\frac{3}{5}$ mean?** Solution: A whole is split into 5 equal parts and 3 are selected: **three fifths**.
2. **Complete $\frac{1}{2} = \frac{?}{8}$.** Solution: Multiply numerator and denominator by 4: **$\frac{1}{2} = \frac{4}{8}$** .
3. **Simplify $\frac{6}{10}$.** Solution: Divide both parts by their common factor 2: **$\frac{6}{10} = \frac{3}{5}$** .
4. **What fraction of 12 counters is 9?** Solution: 9 selected out of 12 gives $\frac{9}{12}$; divide by 3 to get **$\frac{3}{4}$** .
5. **Place $\frac{2}{3}$ on a number line from 0 to 1.** Solution: Split the interval into 3 equal parts and mark the second point: **$\frac{2}{3}$** , between $\frac{1}{2}$ and 1.

N9. Comparing and operating with fractions

Compare fractions by using common denominators, benchmark values, or cross-products. Add or subtract fractions with common denominators by operating on numerators; keep the denominator.

1. **Which is greater: $\frac{3}{4}$ or $\frac{5}{8}$?** Solution: Convert $\frac{3}{4}$ to $\frac{6}{8}$; $\frac{6}{8} > \frac{5}{8}$, so **$\frac{3}{4}$** is greater.
2. **$\frac{1}{6} + \frac{4}{6}$.** Solution: Add numerators: $(1 + 4)/6 = \frac{5}{6}$.
3. **$\frac{7}{8} - \frac{3}{8}$.** Solution: $(7 - 3)/8 = \frac{4}{8} = \frac{1}{2}$.
4. **$\frac{2}{3} + \frac{1}{6}$.** Solution: $\frac{2}{3} = \frac{4}{6}$; $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$.
5. **What is $\frac{3}{4}$ of 20?** Solution: $20 \div 4 = 5$, and $5 \times 3 = 15$.

N10. Decimal place value and operations

Decimals extend place value to tenths, hundredths, and thousandths. Align decimal points when adding or subtracting; compare digits from left to right.

1. **What is the value of 6 in 4.263?** Solution: It is in the hundredths place, so its value is **0.06**.
2. **Compare 0.7 and 0.65.** Solution: Write 0.70; 70 hundredths $>$ 65 hundredths, so **0.7 $>$ 0.65**.
3. **$3.45 + 1.8$.** Solution: Write 1.80, then add: $3.45 + 1.80 = 5.25$.
4. **$10 - 2.76$.** Solution: $10.00 - 2.76 = 7.24$.
5. **Round 6.384 to the nearest tenth.** Solution: The hundredths digit is 8, so round 6.3 up to **6.4**.

N11. Ratios, rates, and percentages

A ratio compares quantities. A rate compares quantities with different units. A percent is a ratio out of 100; use equivalent fractions or decimal multiplication.

1. **There are 2 red marbles and 3 blue marbles. What is the red-to-blue ratio?** Solution: Red:blue = **2:3**.
2. **A car travels 180 miles in 3 hours. What is its unit rate?** Solution: $180 \div 3 = 60$ miles per hour.
3. **Find 25% of 40.** Solution: $25\% = \frac{1}{4}$; $40 \div 4 = 10$.
4. **A recipe uses 2 cups of rice for 5 servings. How much for 10 servings?** Solution: Double the servings, so double the rice: **4 cups**.
5. **Write 0.6 as a percent.** Solution: $0.6 = 60/100$, so it is **60%**.

N12. Patterns and introductory variables

A pattern follows a rule. A variable stands for an unknown or changing value. Use a table, expression, or equation to describe how quantities relate.

1. **Continue 4, 9, 14, 19, ____.** Solution: Add 5 each time; the next number is **24**.
2. **Find the rule for $3 \rightarrow 7$, $5 \rightarrow 11$, $8 \rightarrow 17$.** Solution: Each output is twice the input plus 1: **$y = 2x + 1$** .
3. **If $n = 6$, find $n + 9$.** Solution: Substitute 6: $6 + 9 = 15$.
4. **Solve $x + 14 = 31$.** Solution: Subtract 14 from both sides: **$x = 17$** .
5. **A pattern starts 2, 4, 8, 16. What are the next two terms?** Solution: Double each term: **32, 64**.

N13. Measurement and unit conversion

Measure length, mass, capacity, and temperature with suitable units. Within the metric system, each step changes by a factor of 10, 100, or 1,000; convert before combining unlike units.

1. **How many centimeters are 3 meters?** Solution: $1 \text{ m} = 100 \text{ cm}$, so $3 \times 100 = 300 \text{ cm}$.

2. **Convert 4.5 liters to milliliters.** Solution: $1 \text{ L} = 1,000 \text{ mL}$; $4.5 \times 1,000 = \mathbf{4,500 \text{ mL}}$.
3. **Which is heavier: 2 kg or 1,500 g?** Solution: $2 \text{ kg} = 2,000 \text{ g}$, so **2 kg** is heavier.
4. **A ribbon is 2 m 35 cm long. How many centimeters?** Solution: $2 \text{ m} = 200 \text{ cm}$; $200 + 35 = \mathbf{235 \text{ cm}}$.
5. **A temperature rises from 18°C to 25°C . By how much?** Solution: $25 - 18 = \mathbf{7^{\circ}\text{C}}$.

N14. Time and money

Time uses units related by 60 seconds per minute and 60 minutes per hour. Money combines dollars and cents; align decimal points when adding prices and calculating change.

1. **How many minutes are in 2 hours 15 minutes?** Solution: $2 \times 60 + 15 = \mathbf{135 \text{ minutes}}$.
2. **A movie starts at 6:40 and lasts 1 hour 35 minutes. When does it end?** Solution: $6:40 + 1 \text{ hour} = 7:40$; $+35 \text{ minutes} = \mathbf{8:15}$.
3. **Add \$3.75 and \$2.60.** Solution: $\$3.75 + \$2.60 = \mathbf{\$6.35}$.
4. **Find the change from \$10 for an item costing \$6.48.** Solution: $\$10.00 - \$6.48 = \mathbf{\$3.52}$.
5. **How many quarters equal \$2.00?** Solution: $\$2.00 = 200 \text{ cents}$; $200 \div 25 = \mathbf{8 \text{ quarters}}$.

N15. Shapes, lines, angles, and symmetry

Classify shapes by sides, vertices, parallel or perpendicular lines, and angles. A right angle is 90° . A line of symmetry divides a figure into matching mirror halves.

1. **How many sides does a hexagon have?** Solution: "Hexa-" means six, so a hexagon has **6 sides**.
2. **Classify an angle of 45° .** Solution: It is less than 90° , so it is an **acute angle**.
3. **A rectangle is 9 cm by 4 cm. Are opposite sides equal?** Solution: Yes; each pair of opposite sides has equal length: **9 cm and 4 cm**.
4. **How many lines of symmetry does a square have?** Solution: Two diagonals and two lines through opposite side midpoints give **4 lines**.
5. **Name a 3-D shape with two circular bases.** Solution: A **cylinder** has two parallel circular bases.

N16. Perimeter, area, and volume

Perimeter is distance around a 2-D figure. Area is the number of square units covering it. Volume is the number of cubic units filling a rectangular solid.

1. **Perimeter of a rectangle 8 cm by 3 cm.** Solution: $2 \times (8 + 3) = \mathbf{22 \text{ cm}}$.
2. **Area of a rectangle 8 cm by 3 cm.** Solution: $8 \times 3 = \mathbf{24 \text{ cm}^2}$.
3. **Area of a triangle with base 10 m and height 4 m.** Solution: $(10 \times 4) \div 2 = \mathbf{20 \text{ m}^2}$.
4. **A square has area 49 in^2 . Find its side length.** Solution: $7 \times 7 = 49$, so the side is **7 in**.
5. **Volume of a box 5 cm long, 3 cm wide, and 2 cm high.** Solution: $5 \times 3 \times 2 = \mathbf{30 \text{ cm}^3}$.

N17. Coordinates and transformations

The coordinate plane uses an ordered pair (x, y): move horizontally x first, then vertically y. Translations slide figures; reflections flip them; rotations turn them.

1. **What is the x-coordinate of (4, 7)?** Solution: The first number is x, so it is **4**.
2. **Move (2, 3) two units right.** Solution: Increase x by 2: $(2 + 2, 3) = \mathbf{(4, 3)}$.
3. **Reflect (5, 1) across the y-axis.** Solution: Change the sign of x: **(-5, 1)**.

4. **A rectangle's vertices are (1, 1), (5, 1), (5, 3), and (1, 3). What is its width?** Solution: Horizontal distance is $5 - 1 = 4$ units.
5. **Rotate the arrow pointing up a quarter-turn clockwise. Which way does it point?** Solution: A quarter-turn clockwise from up points **right**.

N18. Tables, graphs, and data questions

Tables organize values. Bar graphs compare categories, line plots show distributions, and line graphs show change. Read titles, labels, units, and scales before interpreting.

1. **A table records 12, 18, and 15 books read Monday–Wednesday. How many total?** Solution: $12 + 18 + 15 = 45$ books.
2. **A bar graph's bars are 7, 4, and 9. Which category is greatest?** Solution: The bar at **9** is tallest, so that category is greatest.
3. **Temperatures are 62° , 65° , 63° , and 70° . What is the increase from first to last?** Solution: $70 - 62 = 8^\circ$.
4. **A line plot has five data points at 2 and three at 3. How many points?** Solution: $5 + 3 = 8$ points.
5. **A pictograph uses one icon for 4 votes. Three icons represent how many votes?** Solution: $3 \times 4 = 12$ votes.

N19. Mean, median, mode, and range

The mean is the sum divided by the number of values. The median is the middle after sorting. The mode is the most frequent value. The range is maximum minus minimum.

1. **Find the mean of 4, 6, and 8.** Solution: $(4 + 6 + 8) \div 3 = 18 \div 3 = 6$.
2. **Find the median of 9, 2, 7, 5, and 6.** Solution: Sorted values are 2, 5, 6, 7, 9; middle value is **6**.
3. **Find the mode of 3, 4, 4, 5, 6, 4.** Solution: 4 occurs most often, so the mode is **4**.
4. **Find the range of 12, 19, 15, and 11.** Solution: $19 - 11 = 8$.
5. **The mean of four scores is 7. Three scores are 5, 8, and 6. Find the fourth.** Solution: Total must be $4 \times 7 = 28$; known total is 19; fourth = $28 - 19 = 9$.

N20. Probability and mathematical communication

Probability describes how likely an event is, from impossible (0) to certain (1). Explain a strategy with labels, equations, units, and a clear conclusion so another reader can check it.

1. **A fair coin is tossed. Probability of heads?** Solution: One favorable outcome out of two: **$1/2$** .
2. **A bag has 3 red and 7 blue counters. Probability of red?** Solution: 3 favorable out of 10 total: **$3/10$** .
3. **What is the probability of rolling a number greater than 6 on a standard die?** Solution: No face is greater than 6, so it is **0 (impossible)**.
4. **A spinner has four equal sections labeled A, B, C, D. Probability of not landing on A?** Solution: Three of four sections are not A: **$3/4$** .
5. **Explain why $4 \times 6 = 24$.** Solution: 4×6 means four equal groups of six; $6 + 6 + 6 + 6 = 24$, so the product is **24**.